PX-200 peristaltic dosing pump

Service manual · Edition 03/2028 · Original instructions

Aldermoor Fluid Systems

Synthetic sample: the product, the manufacturer and all values in this manual are invented. Do not use it for real equipment.

1 Safety

1.1 About this manual

This manual describes the installation, operation and maintenance of the PX-200 series of peristaltic dosing pumps. Read it completely before you install or operate the pump, and keep it near the pump for later reference. The menus described here refer to firmware version 2.4 and later.

1.2 Signal words

- **DANGER** marks a hazard that will cause death or serious injury if it is not avoided.
- **WARNING** marks a hazard that can cause death or serious injury.
- **CAUTION** marks a hazard that can cause minor injury or damage to the pump.
- **NOTE** gives information that makes the work easier.

1.3 Intended use

The PX-200 is intended for dosing liquid chemicals in water treatment, cooling water and process applications. The pump may be used only with liquids that are compatible with the hose material and the pump head, and only within the limits given in the technical data. Any other use is not intended and may cause injury or damage.

1.4 General safety instructions

- Disconnect the mains supply and secure it against reconnection before you open the terminal box or the pump head.
- Do not operate the pump with the pump head cover removed: the rotor can trap fingers.
- Check the safety data sheet of the dosed liquid and wear the protective equipment it requires.
- Relieve the pressure in the discharge line before you disconnect it.
- Collect leaking liquid and dispose of it according to local regulations.
- Use only original spare parts. Other parts can affect the safety of the pump and void the warranty.
- Do not modify the pump or its software. Modifications require written approval from the manufacturer.

> **WARNING** Wear a face shield when you open the discharge line, even after the pressure has been relieved: liquid can still spray from the hose or the connector. Keep an eye-wash bottle within reach of the pump, and rinse splashes on the skin or in the eyes at once with plenty of clean water. Report every such incident to the person responsible for the installation.

1.5 Personnel

Installation, electrical connection and maintenance must be carried out by trained specialists who are familiar with this manual and with the regulations that apply on site. Operators must be instructed in the use of the pump and in the hazards of the dosed liquid.

1.6 Residual risks

Even when the pump is used as intended, some risks remain: chemicals can escape if a hose bursts, the motor surface can become hot after long operation at full speed, and the pump can start unexpectedly on a remote signal. Install a leak sensor and an external stop where these risks are not acceptable.

1.7 Warranty

The warranty covers defects in material and workmanship for 24 months from delivery. It does not cover wear parts such as hoses, rollers and injection valve seals, or damage caused by unsuitable liquids, incorrect installation or the use of non-original parts. Warranty claims must be made in writing and must include the serial number and a description of the fault.

1.8 Labels on the pump

Label	Position	Meaning
Rotating parts	pump head cover	Do not open the cover while the pump is running.
Hot surface	motor housing	The surface can exceed 60 °C after long operation.
Read the manual	controller cover	Read this manual before installation and maintenance.
Direction of flow	pump head, both connectors	Arrow from the suction side to the discharge side.
Earth terminal	terminal box	Connection point for the protective earth conductor.
Rating plate	right-hand side	Model, serial number and electrical data.

Replace labels that are damaged or no longer legible. Replacement labels are part of the service kit.

2 Product description

2.1 Overview

The PX-200 is a peristaltic pump: a rotor with three rollers squeezes a flexible hose against the curved track of the pump head and pushes the liquid forward. The liquid touches only the hose and the connectors, so the pump has no valves or seals in contact with the liquid. It can run dry without damage and can pump liquids that contain gas bubbles or solids.

2.2 Models

Model	Flow (l/h)	Max. pressure (bar)	Hose ID (mm)	Motor (W)
PX-200-S	0.1–12	7	1.6 / 3.2	40
PX-200-M	0.4–48	7	4.8	60
PX-200-L	1.0–120	4	6.4	90
PX-200-XL	2.0–230	3	8.0	120
PX-200-HP	0.2–20	12	3.2	90
PX-200-EX	0.4–48	7	4.8	60

The PX-200-EX has a conductive housing and is intended for areas where explosive atmospheres can occur. Its additional instructions are given in a separate supplement and take precedence over this manual.

2.3 Pump head

The pump head consists of the housing with the hose track, the rotor with three rollers, the hose element and the transparent cover. The cover is fixed with four screws and has a safety switch that stops the motor when the cover is opened. The hose element is supplied ready for installation, with both connectors pressed on.

2.4 Drive and controller

The drive consists of a brushless DC motor and a two-stage planetary gearbox. The controller adjusts the motor speed between 0.5 and 100 % of the nominal speed and counts the revolutions of the rotor, so that the dosed volume can be calculated without a flow meter. A calibration value is stored for each hose size.

2.5 Rating plate

The rating plate is on the right-hand side of the housing. It shows:

- model and serial number;
- year of manufacture;

- supply voltage and frequency;
- power consumption;
- maximum discharge pressure;
- degree of protection.

Quote the model and the serial number in all enquiries and spare part orders.

2.6 Scope of delivery

- pump with controller and mains cable;
- hose element installed in the pump head;
- wall bracket with locking screw and drilling template;
- connector set for the suction and discharge lines;
- leak sensor with cable;
- hose lever and union nut tool;
- this manual and a quick reference card.

Check the delivery for completeness and for transport damage as soon as it arrives. Report any damage to the carrier and to your distributor within three working days.

2.7 Dimensions

Model	Width (mm)	Height (mm)	Depth (mm)	Weight (kg)
PX-200-S	190	240	170	4.8
PX-200-M	205	250	185	5.6
PX-200-L	225	265	205	6.7
PX-200-XL	240	280	220	7.9
PX-200-HP	205	250	200	6.1
PX-200-EX	205	250	185	6.0

3 Installation

3.1 Location

Install the pump in a dry, frost-free place that is protected from direct sunlight. The ambient temperature must be between 0 and 45 °C.

- Leave at least 30 cm of free space in front of the pump head for hose changes.
- Install the pump below the level of the storage tank where possible, so that the suction line stays filled.
- Provide a drip tray or bund under the pump and the storage tank.
- Keep the pump away from heat sources and from vibrating machines.

3.2 Mounting

The pump can be mounted on a wall bracket, on a console or directly on the lid of a storage tank.

1. Mark the four fixing holes with the drilling template supplied.
2. Drill the holes and insert suitable wall plugs.
3. Fix the bracket with four M6 screws.
4. Hang the pump on the bracket and secure it with the locking screw.
5. Check that the pump head is horizontal and easy to reach.

3.3 Suction and discharge lines

Use hoses or pipes whose material is compatible with the dosed liquid. The inner diameter of the suction line must be at least as large as that of the pump hose, and the suction line should be as short as possible.

Install a pressure relief valve in the discharge line if the line can be closed while the pump is running, for example by a manual valve or a blocked injection nozzle. Without such a valve, the pressure can rise until the hose bursts.

Fit the injection valve at the dosing point. It prevents the dosing line from draining and keeps liquid from the main line out of the dosing line.

3.4 Electrical connection

The pump is supplied with a 2 m mains cable with a plug. Signal and alarm cables are connected in the terminal box under the controller cover.

Terminal	Function	Rating
X1.1–X1.3	Mains supply L, N, PE	100–240 V AC, 50/60 Hz
X2.1–X2.2	Pulse input	5–30 V DC, max. 100 Hz
X2.3–X2.4	Analogue input	0/4–20 mA, 250 Ω
X2.5–X2.6	External stop	potential-free contact
X3.1–X3.2	Level switch, pre-alarm	potential-free contact
X3.3–X3.4	Level switch, empty	potential-free contact
X4.1–X4.3	Alarm relay	250 V AC, 2 A, changeover
X4.4–X4.5	Analogue output	4–20 mA, max. 500 Ω
X5.1–X5.2	Leak sensor	supplied with the pump
X6	Service interface	USB-C

Lay signal cables separately from mains cables. Use screened cable for the analogue input and output, and connect the screen at the pump end only.

3.5 Signal inputs

In pulse mode, the pump delivers a set volume for each pulse from a water meter. In analogue mode, the flow is proportional to the input current; the range is set with parameters P20 to P23. The external stop input has priority over all operating modes.

3.6 Installation examples

Dosing into a pressurised pipe. Fit the injection valve on top of the pipe or at the side, never at the bottom, where sediment collects. The opening pressure of the injection valve must be higher than the pressure difference

between the tank and the dosing point; otherwise liquid can siphon through the pump when it is stopped. In pipes with changing pressure, add a pressure-holding valve close to the pump.

Dosing into an open tank. Lead the discharge line into the tank from above and end it below the lowest liquid level, so that the dosed liquid mixes quickly. Fix the end of the line so that it cannot move out of the liquid when the mixer runs. An injection valve is not needed, but a non-return valve prevents the line from draining.

Dosing with a pulse water meter. Connect the meter to the pulse input and select pulse mode. The pulse volume is calculated from the meter constant and the required dose. Example: the meter gives one pulse per 10 litres and the required dose is 2 ml of product per 100 litres of water; set the pulse volume P11 to 0.2 ml.

Suction from a drum. Use a suction lance with a foot valve and a level switch, and place the drum in a bund. Keep the suction height below 3 m for water-like liquids and below 1.5 m for liquids that release gas. Replace the drum only when the pump has stopped and the “tank empty” message is shown.

4 Commissioning

1. Check that the supply voltage matches the rating plate.
2. Check that all hose connections are tight and that the valves in the suction and discharge lines are open.
3. Switch on the supply. The display shows the firmware version and then the main screen.
4. Select the language and the units in the setup menu.
5. Select the installed hose size in parameter P05.
6. Prime the pump: press and hold the PRIME key until liquid flows from the discharge line without bubbles.
7. Calibrate the pump as described in section 6.
8. Set the operating mode and the dosing rate.

9. Check the installation for leaks during the first hour of operation.
10. Record the commissioning data in the maintenance record at the end of this manual.

If the pump does not deliver, check the direction of rotation shown on the display and the position of the hose in the track.

4.1 Commissioning record

Copy this table into the maintenance record or the plant documentation.

Item	Value
Date of commissioning	
Serial number	
Firmware version	
Hose size and material	
Calibration factor (P31)	
Operating mode (P06)	
Dosing rate or pulse volume	
Commissioned by	

5 Operation

5.1 Control panel

The control panel has a backlit display and six keys.

Key	Function
START/STOP	Starts or stops dosing in all modes
PRIME	Runs the pump at full speed while the key is pressed
MENU	Opens the parameter menu; pressed again, returns to the main screen

Key	Function
UP / DOWN	Change the selected value
OK	Confirms a value or opens the selected menu item

5.2 Operating modes

Manual mode. The pump runs continuously at the set rate in ml/min or l/h.

Pulse mode. Each input pulse starts the delivery of a set volume. Pulses that arrive during a delivery are stored, up to 99 pulses.

Analogue mode. The flow follows the input current. Below the lower point of the range the pump stops; above the upper point it runs at the flow set for the upper point.

Batch mode. The pump delivers a set volume and stops. A new batch is started with the START key or with a pulse at the pulse input.

Timer mode. Up to eight batches per day are started at set times. The internal clock keeps the time for at least five days without supply.

5.3 Parameter list

Code	Parameter	Range	Default
P01	Language	English, Deutsch, Français, Español	English
P02	Units	ml/min, l/h, gal/h	l/h
P03	Display contrast	1–10	6
P04	Access code	0000–9999	0000
P05	Hose size	1.6, 3.2, 4.8, 6.4, 8.0 mm	4.8 mm
P06	Operating mode	manual, pulse, analogue, batch, timer	manual
P07	Manual rate	0.5–100 %	50 %
P08	Maximum speed	10–100 %	100 %
P09	Ramp time	0–10 s	2 s
P10	Direction of rotation	clockwise, anticlockwise	clockwise
P11	Pulse volume	0.1–999 ml	10 ml

Code	Parameter	Range	Default
P12	Pulse memory	on, off	on
P13	Pulse divider	1–99	1
P14	Batch volume	0.01–999 l	1.00 l
P15	Batch start	key, pulse, timer	key
P16	Timer batches per day	0–8	0
P17	Anti-drip reverse	0–5 revolutions	0
P18	Leak sensor	on, off	on
P19	Level input logic	NO, NC	NC
P20	Analogue input range	0–20 mA, 4–20 mA	4–20 mA
P21	Analogue lower point	0–20 mA	4.0 mA
P22	Analogue upper point	0–20 mA	20.0 mA
P23	Flow at upper point	0–100 %	100 %
P24	Analogue output source	speed, flow, alarm	flow
P25	Alarm relay function	fault, empty, pre-alarm, all	all
P26	Alarm relay logic	energised, de-energised	energised
P27	External stop logic	NO, NC	NO
P28	Restart after power failure	automatic, manual	manual
P29	Hose run time warning	0–5000 h	1500 h
P30	Service interval	0–8760 h	4000 h
P31	Calibration factor	0.50–1.50	1.00
P32	Total volume counter	read only	–
P33	Hose run time counter	read only	–
P34	Fault history	read only	–
P35	Factory reset	yes, no	no
P36	Firmware version	read only	–

Parameters P04, P10 and P31 to P35 are only shown after the access code has been entered.

5.4 Menu structure

Menu	Contents
Setup	language, units, display contrast, access code, hose size
Operation	operating mode, rates, volumes, batch and timer settings
Inputs	pulse input, analogue input, external stop, level switches, leak sensor
Outputs	alarm relay, analogue output
Service	calibration, counters, firmware update, factory reset
Info	model, serial number, firmware version, fault history

Use UP and DOWN to move between menu items and OK to open an item. The menu closes automatically when no key is pressed for two minutes; values that have not been confirmed with OK are discarded.

5.5 Counters and messages

The controller keeps three counters: the total volume dosed since commissioning, the run time of the installed hose and the time since the last service. The total volume counter cannot be reset. The hose run time counter is reset after a hose change, and the service counter after a service. When a counter reaches the limit set in P29 or P30, the display shows a warning and the alarm relay switches if it is set to “all”.

The last 50 faults and warnings are stored with date and time in the fault history (P34). The history is kept during a firmware update and after a power failure.

5.6 Remote monitoring

The analogue output gives a signal of 4–20 mA that follows the speed, the flow or the alarm state, as set in P24. With the setting “alarm”, the output is 4 mA in normal operation, 12 mA when a warning is active and 20 mA when a fault has stopped the pump. Together with the alarm relay this allows a control system to monitor the pump without a data connection.

6 Flow calibration

The controller converts the rotor speed into a flow rate with a calibration value for each hose size. The factory values are typical values for new hoses at 20 °C and a discharge pressure of 1 bar. Calibrate the pump after commissioning, after every hose change and whenever the dosed liquid changes.

1. Fill a measuring cylinder with the dosed liquid and place the end of the suction line in it.
2. Start the calibration in the menu Service › Calibration.
3. Let the pump run for at least 60 seconds, or until at least 100 ml have been drawn.
4. Enter the volume drawn from the cylinder.
5. Confirm the new calibration factor with OK.

Typical flow in ml/min at 1 bar, by hose inner diameter and speed (% of nominal speed):

Hose	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %	100 %
1.6 mm	8.0	16.0	24.0	32.0	40.0	48.0	56.0	64.0	72.0	80.0
3.2 mm	31.0	62.0	93.0	124	155	186	217	248	279	310
4.8 mm	69.0	138	207	276	345	414	483	552	621	690
6.4 mm	122	244	366	488	610	732	854	976	1098	1220
8.0 mm	189	378	567	756	945	1134	1323	1512	1701	1890

Higher discharge pressure, viscous liquids and worn hoses reduce the flow. With liquids that release gas, such as sodium hypochlorite, the flow can be up to 15 % lower than the table value.

7 Hose life

The life of the pump hose depends on the speed, the discharge pressure, the temperature and the dosed liquid. The tables give typical operating hours with

water at 20 °C for TPE hoses. Silicone hoses reach about half of these values; FKM hoses about two thirds.

Typical hose life in hours at 1 bar, by speed:

Hose	25 %	50 %	75 %	100 %
1.6 mm	6820	4200	3160	2590
3.2 mm	6210	3820	2880	2350
4.8 mm	5590	3440	2590	2120
6.4 mm	4970	3060	2300	1880
8.0 mm	4350	2680	2020	1650

Typical hose life in hours at 4 bar, by speed:

Hose	25 %	50 %	75 %	100 %
1.6 mm	3750	2310	1740	1420
3.2 mm	3410	2100	1580	1290
4.8 mm	3070	1890	1420	1160
6.4 mm	2730	1680	1270	1040
8.0 mm	2390	1470	1110	910

Replace the hose at the latest when the hose run time warning appears, or earlier if the flow drops by more than 10 % after a calibration.

7.1 Hose materials

Material	Temperature	Max. pressure	Typical use
TPE	0 to 50 °C	7 bar	standard hose for water treatment
Silicone	−20 to 80 °C	2 bar	low-pressure applications
FKM	0 to 60 °C	5 bar	oils and the solvents listed in the supplement
EPDM	0 to 60 °C	4 bar	alkaline cleaning solutions

Check the compatibility of each liquid in the chemical resistance supplement before you select the hose material. If in doubt, ask the distributor for a test

hose and check it for swelling after one week of operation.

8 Maintenance

8.1 Schedule

Task	Interval
Check the pump and the lines for leaks	daily
Compare the dosed volume on the display with the tank level	weekly
Clean the pump head cover	monthly
Check the hose for swelling, cracks and flat spots	monthly
Recalibrate the pump	after each hose change
Replace the hose element	at the hose run time warning, at the latest every 12 months
Inspect the rollers and the rotor bearings	at each hose change
Clean the hose track	at each hose change
Check the leak sensor	every 6 months
Check the injection valve	every 6 months
Replace the injection valve seals	every 12 months
Check the terminal box for moisture and loose wires	every 12 months
Replace the rotor	every 8,000 operating hours
Replace the gearbox	every 20,000 operating hours

8.2 Maintenance by the operator

The daily, weekly and monthly checks can be carried out by instructed operators. They do not require the pump head or the controller to be opened. Record any leak, unusual noise or deviation between the dosed volume and the tank level, and report it to the person responsible for maintenance.

8.3 Maintenance by service personnel

Hose changes, inspections of the pump head, work on the injection valve and all work on the electrical installation are carried out by trained service personnel. The intervals in the schedule apply to continuous operation at up to 50 % speed; at higher speeds, or with abrasive or crystallising liquids, shorten them according to experience.

8.4 Service kits

Kit	Contents	Used for
Hose kit	hose element, connector seals, silicone grease	hose change (9.1)
Roller kit	three rollers with bearings, retaining clips	roller replacement (9.2)
Valve kit	O-rings, valve ball, sealing tape	injection valve service (9.6)
Annual kit	hose kit, valve kit, controller cover seal, fuse set	annual service
Rotor kit	rotor with rollers, rotor fixing screw, shaft seal	rotor replacement (9.4)

9 Maintenance procedures

> **WARNING** Stop the pump and disconnect the mains supply before every maintenance task, unless the procedure says otherwise.

9.1 Replacing the hose element

1. Stop the pump and disconnect the mains supply.
2. Close the valves in the suction and discharge lines and relieve the pressure.
3. Remove the four screws of the pump head cover and take off the cover.
4. Unscrew the union nuts of the hose connectors.
5. Turn the rotor by hand with the hose lever and lift the old hose out of the track.
6. Clean the hose track and inspect the rollers (9.2 and 9.3).

7. Insert the new hose element, starting at the suction side, and turn the rotor by hand until the hose lies evenly in the track.
8. Tighten the union nuts by hand and then a quarter turn with the tool supplied.
9. Fit the cover and tighten the screws to the torque given in the technical data.
10. Reset the hose run time counter (P33) and recalibrate the pump.

9.2 Inspecting the rollers

1. Remove the pump head cover.
2. Turn each roller by hand. It must turn freely, without noise or play.
3. Check the running surface for flat spots, cracks and deposits.
4. Replace the complete roller set if one roller is damaged; do not replace single rollers.
5. Fit the cover.

9.3 Cleaning the hose track

1. Remove the hose element.
2. Wipe the track with a cloth moistened with water. Remove deposits of dried chemical with a plastic scraper; do not use metal tools.
3. Dry the track completely before you insert the new hose.
4. Apply a thin film of the silicone grease supplied to the track.

9.4 Replacing the rotor

1. Remove the pump head cover and the hose element.
2. Hold the rotor and loosen the rotor fixing screw in the centre.
3. Pull the rotor off the shaft. If it is stuck, use the puller from the service kit.
4. Clean the shaft and check the shaft seal for leaks.
5. Push the new rotor onto the shaft until it stops, with the flat on the shaft aligned with the flat in the rotor hub.
6. Tighten the rotor fixing screw to the torque given in the technical data.

7. Insert the hose element and fit the cover.

9.5 Checking the leak sensor

The leak sensor is a conductive probe at the lowest point of the pump head.

1. Stop the pump.
2. Lift the sensor out of its holder and dip the tip into a cup of tap water.
3. The display must show the leak alarm, and the alarm relay must switch.
4. Dry the sensor, put it back in the holder and acknowledge the alarm with OK.

9.6 Servicing the injection valve

1. Relieve the pressure in the discharge line.
2. Unscrew the injection valve from the main line.
3. Dismantle the valve and clean all parts in water.
4. Replace the O-rings and the valve ball from the seal kit.
5. Assemble the valve and check that the ball moves freely.
6. Refit the valve with new sealing tape on the thread.

9.7 Replacing the controller fuse

1. Disconnect the mains supply.
2. Open the controller cover (four screws).
3. Remove the fuse holder cap and replace the fuse with one of the same type.
4. Close the cover and check that the seal is seated correctly.

9.8 Updating the firmware

1. Download the firmware file from the manufacturer’s service portal.
2. Connect a computer to the service interface X6 with a USB-C cable.
3. Start the update in the menu Service › Firmware. The pump stops during the update.

4. Do not disconnect the supply until the display shows the main screen again.
5. Check the parameter settings; they are kept during the update.

9.9 Replacing the gearbox

1. Disconnect the mains supply and remove the pump from the wall bracket.
2. Remove the pump head cover, the hose element and the rotor (9.4).
3. Remove the four screws of the pump head housing and lift the housing off.
4. Remove the four gearbox flange screws and pull the gearbox off the motor shaft.
5. Check the motor shaft and the coupling for wear.
6. Fit the new gearbox, aligning the coupling, and tighten the flange screws to the torque given in the technical data.
7. Refit the pump head housing, the rotor and the hose element.
8. Run the pump for ten minutes at full speed and check for noise and leaks.

9.10 Replacing the controller board

1. Note the parameter settings, or save them to a computer through the service interface.
2. Disconnect the mains supply and wait two minutes.
3. Open the controller cover and disconnect all plugs from the board.
4. Remove the five board screws and lift out the board, holding it by the edges.
5. Fit the new board and reconnect the plugs; each plug fits only one socket.
6. Close the cover and switch on the supply.
7. Enter the parameter settings again and recalibrate the pump.

9.11 Cleaning the pump

1. Stop the pump.

2. Wipe the housing and the display with a damp cloth. Do not use solvents or abrasive cleaners.
3. Clean the pump head cover with water and a mild detergent.
4. Check that the ventilation openings of the motor are free.
5. Remove dried chemical from the connectors with a soft brush.

9.12 Storing spare hoses

Store spare hose elements in their original bag in a dark, dry place at 5 to 25 °C. Do not store them near ozone sources such as electric motors or welding equipment. Hoses older than three years should not be installed, even if they have never been used.

10 Fault finding

Faults (E) stop the pump and switch the alarm relay. Warnings (W) are shown on the display, but the pump continues to run. Acknowledge a message with OK after the cause has been removed.

Display or symptom	Possible cause	Remedy
No display	No supply, or the controller fuse has blown	Check the supply; replace the fuse (9.7)
E01 Cover open	Pump head cover not closed	Close the cover and tighten the screws
E02 Motor blocked	Rotor blocked by a foreign body or a twisted hose	Stop the pump; check the hose and the rotor
E03 Overcurrent	Discharge pressure too high	Check the discharge line and the injection valve for blockage
E04 Overtemperature	Ambient temperature too high, or the motor is covered	Improve the ventilation; let the motor cool down
E05 Leak	Liquid at the leak sensor	Find and repair the leak; dry the sensor (9.5)
E06 Tank empty	Level switch “empty” active	Refill the tank
W07 Tank low	Level switch “pre-alarm” active	Prepare a refill

Display or symptom	Possible cause	Remedy
E08 Analogue input	Input current below 3.5 mA with the range 4–20 mA	Check the signal cable and the signal source
W09 Pulse overflow	More than 99 pulses stored	Increase the pulse volume or the pulse divider
W10 Hose run time	Hose run time warning reached	Replace the hose element (9.1)
W11 Service due	Service interval reached	Carry out the maintenance tasks and reset the counter
E12 Memory	Parameter memory faulty	Reset to factory settings; contact service if the fault returns
E13 Speed sensor	No signal from the motor speed sensor	Check the motor cable; replace the motor
E14 External stop	External stop input active	Check the external stop circuit
Pump runs, no flow	Hose not seated, suction line leaking or a valve closed	Check the hose position, the suction line and the valves
Flow too low	Worn hose, out-of-date calibration or viscous liquid	Replace the hose; recalibrate the pump
Flow too high	Siphoning through the pump at low counter-pressure	Fit an injection valve or a pressure-holding valve
Hose bursts early	Discharge pressure too high, or unsuitable hose material	Check the pressure and the hose material
Liquid in the pump head	Hose burst or connector leaking	Stop the pump; clean the head; replace the hose
Noise from the pump head	Roller bearings worn or the track is dry	Inspect the rollers (9.2); clean the track (9.3)
Wrong display language	Parameter P01 changed	Set P01 to the required language

Before you contact service, note the model, the serial number, the firmware version and the last entries of the fault history. Check that the fault is not caused by the installation, for example a closed valve or a blocked injection valve.

11 Spare parts and accessories

11.1 Spare parts

Order spare parts from your local distributor and quote the part number, the model and the serial number of the pump. Parts marked “yes” in the last column should be kept in stock on site.

Part no.	Description	Model	Stock
PX2-1104	Hose element, TPE, ID 1.6 / 3.2 mm	PX-200-S	yes
PX2-1111	Hose element, silicone, ID 1.6 / 3.2 mm	PX-200-S	–
PX2-1118	Hose element, FKM, ID 1.6 / 3.2 mm	PX-200-S	–
PX2-1125	Rotor, complete with three rollers	PX-200-S	–
PX2-1132	Roller set (3 rollers with bearings)	PX-200-S	yes
PX2-1139	Pump head cover with screws	PX-200-S	–
PX2-1146	Cover safety switch	PX-200-S	–
PX2-1153	Connector set, PVDF	PX-200-S	yes
PX2-1160	Connector set, PP	PX-200-S	–
PX2-1167	Connector set, stainless steel	PX-200-S	–
PX2-1174	Injection valve, PVDF	PX-200-S	–
PX2-1181	Injection valve, PP	PX-200-S	–
PX2-1188	Seal kit for injection valve	PX-200-S	yes
PX2-1195	Planetary gearbox	PX-200-S	–
PX2-1202	BLDC motor with speed sensor	PX-200-S	–
PX2-1209	Controller board	PX-200-S	–
PX2-1216	Pump head housing with track	PX-200-S	–
PX2-1223	Shaft seal	PX-200-S	yes
PX2-1230	Hose lever	PX-200-S	–
PX2-1237	Controller cover seal	PX-200-S	–
PX2-1244	Hose element, TPE, ID 4.8 mm	PX-200-M	yes

Part no.	Description	Model	Stock
PX2-1251	Hose element, silicone, ID 4.8 mm	PX-200-M	–
PX2-1258	Hose element, FKM, ID 4.8 mm	PX-200-M	–
PX2-1265	Rotor, complete with three rollers	PX-200-M	–
PX2-1272	Roller set (3 rollers with bearings)	PX-200-M	yes
PX2-1279	Pump head cover with screws	PX-200-M	–
PX2-1286	Cover safety switch	PX-200-M	–
PX2-1293	Connector set, PVDF	PX-200-M	yes
PX2-1300	Connector set, PP	PX-200-M	–
PX2-1307	Connector set, stainless steel	PX-200-M	–
PX2-1314	Injection valve, PVDF	PX-200-M	–
PX2-1321	Injection valve, PP	PX-200-M	–
PX2-1328	Seal kit for injection valve	PX-200-M	yes
PX2-1335	Planetary gearbox	PX-200-M	–
PX2-1342	BLDC motor with speed sensor	PX-200-M	–
PX2-1349	Controller board	PX-200-M	–
PX2-1356	Pump head housing with track	PX-200-M	–
PX2-1363	Shaft seal	PX-200-M	yes
PX2-1370	Hose lever	PX-200-M	–
PX2-1377	Controller cover seal	PX-200-M	–
PX2-1384	Hose element, TPE, ID 6.4 mm	PX-200-L	yes
PX2-1391	Hose element, silicone, ID 6.4 mm	PX-200-L	–
PX2-1398	Hose element, FKM, ID 6.4 mm	PX-200-L	–
PX2-1405	Rotor, complete with three rollers	PX-200-L	–
PX2-1412	Roller set (3 rollers with bearings)	PX-200-L	yes
PX2-1419	Pump head cover with screws	PX-200-L	–
PX2-1426	Cover safety switch	PX-200-L	–
PX2-1433	Connector set, PVDF	PX-200-L	yes

Part no.	Description	Model	Stock
PX2-1440	Connector set, PP	PX-200-L	–
PX2-1447	Connector set, stainless steel	PX-200-L	–
PX2-1454	Injection valve, PVDF	PX-200-L	–
PX2-1461	Injection valve, PP	PX-200-L	–
PX2-1468	Seal kit for injection valve	PX-200-L	yes
PX2-1475	Planetary gearbox	PX-200-L	–
PX2-1482	BLDC motor with speed sensor	PX-200-L	–
PX2-1489	Controller board	PX-200-L	–
PX2-1496	Pump head housing with track	PX-200-L	–
PX2-1503	Shaft seal	PX-200-L	yes
PX2-1510	Hose lever	PX-200-L	–
PX2-1517	Controller cover seal	PX-200-L	–
PX2-1524	Hose element, TPE, ID 8.0 mm	PX-200-XL	yes
PX2-1531	Hose element, silicone, ID 8.0 mm	PX-200-XL	–
PX2-1538	Hose element, FKM, ID 8.0 mm	PX-200-XL	–
PX2-1545	Rotor, complete with three rollers	PX-200-XL	–
PX2-1552	Roller set (3 rollers with bearings)	PX-200-XL	yes
PX2-1559	Pump head cover with screws	PX-200-XL	–
PX2-1566	Cover safety switch	PX-200-XL	–
PX2-1573	Connector set, PVDF	PX-200-XL	yes
PX2-1580	Connector set, PP	PX-200-XL	–
PX2-1587	Connector set, stainless steel	PX-200-XL	–
PX2-1594	Injection valve, PVDF	PX-200-XL	–
PX2-1601	Injection valve, PP	PX-200-XL	–
PX2-1608	Seal kit for injection valve	PX-200-XL	yes
PX2-1615	Planetary gearbox	PX-200-XL	–
PX2-1622	BLDC motor with speed sensor	PX-200-XL	–

Part no.	Description	Model	Stock
PX2-1629	Controller board	PX-200-XL	–
PX2-1636	Pump head housing with track	PX-200-XL	–
PX2-1643	Shaft seal	PX-200-XL	yes
PX2-1650	Hose lever	PX-200-XL	–
PX2-1657	Controller cover seal	PX-200-XL	–
PX2-1664	Hose element, TPE, ID 3.2 mm	PX-200-HP	yes
PX2-1671	Hose element, silicone, ID 3.2 mm	PX-200-HP	–
PX2-1678	Hose element, FKM, ID 3.2 mm	PX-200-HP	–
PX2-1685	Rotor, complete with three rollers	PX-200-HP	–
PX2-1692	Roller set (3 rollers with bearings)	PX-200-HP	yes
PX2-1699	Pump head cover with screws	PX-200-HP	–
PX2-1706	Cover safety switch	PX-200-HP	–
PX2-1713	Connector set, PVDF	PX-200-HP	yes
PX2-1720	Connector set, PP	PX-200-HP	–
PX2-1727	Connector set, stainless steel	PX-200-HP	–
PX2-1734	Injection valve, PVDF	PX-200-HP	–
PX2-1741	Injection valve, PP	PX-200-HP	–
PX2-1748	Seal kit for injection valve	PX-200-HP	yes
PX2-1755	Planetary gearbox	PX-200-HP	–
PX2-1762	BLDC motor with speed sensor	PX-200-HP	–
PX2-1769	Controller board	PX-200-HP	–
PX2-1776	Pump head housing with track	PX-200-HP	–
PX2-1783	Shaft seal	PX-200-HP	yes
PX2-1790	Hose lever	PX-200-HP	–
PX2-1797	Controller cover seal	PX-200-HP	–
PX2-1804	Hose element, TPE, ID 4.8 mm	PX-200-EX	yes
PX2-1811	Hose element, silicone, ID 4.8 mm	PX-200-EX	–

Part no.	Description	Model	Stock
PX2-1818	Hose element, FKM, ID 4.8 mm	PX-200-EX	–
PX2-1825	Rotor, complete with three rollers	PX-200-EX	–
PX2-1832	Roller set (3 rollers with bearings)	PX-200-EX	yes
PX2-1839	Pump head cover with screws	PX-200-EX	–
PX2-1846	Cover safety switch	PX-200-EX	–
PX2-1853	Connector set, PVDF	PX-200-EX	yes
PX2-1860	Connector set, PP	PX-200-EX	–
PX2-1867	Connector set, stainless steel	PX-200-EX	–
PX2-1874	Injection valve, PVDF	PX-200-EX	–
PX2-1881	Injection valve, PP	PX-200-EX	–
PX2-1888	Seal kit for injection valve	PX-200-EX	yes
PX2-1895	Planetary gearbox	PX-200-EX	–
PX2-1902	BLDC motor with speed sensor	PX-200-EX	–
PX2-1909	Controller board	PX-200-EX	–
PX2-1916	Pump head housing with track	PX-200-EX	–
PX2-1923	Shaft seal	PX-200-EX	yes
PX2-1930	Hose lever	PX-200-EX	–
PX2-1937	Controller cover seal	PX-200-EX	–
PX2-1944	Display module	all	–
PX2-1951	Keypad foil	all	–
PX2-1958	Fuse set, 5 × T 2 A	all	yes
PX2-1965	Leak sensor with 1 m cable	all	yes
PX2-1972	Mains cable, 2 m, with plug	all	–
PX2-1979	Wall bracket with locking screw	all	–
PX2-1986	Drilling template	all	–
PX2-1993	Silicone grease, 20 g	all	yes

11.2 Accessories

Part no.	Accessory
PX2-7010	Suction lance for 25 l containers, with level switch
PX2-7017	Suction lance for 200 l drums, with level switch
PX2-7024	Foot valve, PVDF, with strainer
PX2-7031	Pressure relief valve, PVDF, 3–10 bar
PX2-7038	Pressure-holding valve, PVDF, 1.5 bar
PX2-7045	Pulsation damper, 0.2 l
PX2-7052	Drip tray, polyethylene, 40 l
PX2-7059	Console for tank lids
PX2-7066	Signal cable, 5 m, screened
PX2-7073	Signal cable, 10 m, screened
PX2-7080	Pulse water meter adapter
PX2-7087	USB-C service cable, 2 m
PX2-7094	Service kit with puller, torque screwdriver bits and hose lever
PX2-7101	Dosing line, PE, 4/6 mm, 50 m
PX2-7108	Dosing line, PTFE, 4/6 mm, 25 m
PX2-7115	Calibration cylinder, 250 ml

12 Technical data

12.1 General

Property	Value
Ambient temperature	0 to 45 °C
Liquid temperature	0 to 50 °C
Storage temperature	−20 to 60 °C
Relative humidity	max. 90 %, non-condensing
Degree of protection	IP65

Property	Value
Weight	4.8 kg (S) to 7.9 kg (XL)
Sound pressure level	below 55 dB(A) at 1 m
Housing	glass-fibre reinforced polypropylene
Display	LCD, 128 × 64 pixels, backlit

12.2 Electrical data

Property	Value
Supply	100–240 V AC, 50/60 Hz
Power consumption	max. 150 W
Controller fuse	T 2 A, 5 × 20 mm
Alarm relay	250 V AC, 2 A
Analogue input	0/4–20 mA, 250 Ω
Pulse input	5–30 V DC, max. 100 Hz

12.3 Tightening torques

Screw	Thread	Torque
Pump head cover screws	M5	3.5 N·m
Rotor fixing screw	M6	9.0 N·m
Gearbox flange screws	M5	4.5 N·m
Wall bracket screws	M6	8.0 N·m
Terminal box cover screws	M4	1.5 N·m
Hose connector union nuts	—	hand-tight plus a quarter turn

Use a calibrated torque screwdriver. Do not use power tools on the plastic parts of the pump head.

13 Decommissioning and disposal

Before the pump is taken out of service for a long period, or sent for repair:

1. Flush the pump with water or a suitable neutral liquid for at least five minutes.
2. Relieve the pressure and empty the suction and discharge lines.
3. Wear butyl gloves and safety goggles when you remove the hose element.
4. Remove the hose element and store it in a dark, dry place.
5. Fill in the decontamination declaration and attach it to the outside of the packaging.

The pump contains electronic components and must not be disposed of with household waste. Return it to the manufacturer or to a collection point for electrical equipment. Hoses that have been in contact with chemicals are disposed of as hazardous waste.

14 Maintenance record

Date	Work carried out	Hose run time (h)	Name

Date	Work carried out	Hose run time (h)	Name